Jichen Feng

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github.com/T3S1AMAX

AI for Science | AI for Math | Computational Chemistry

PROFESSIONAL SUMMARY

Ph.D. candidate in computational chemistry and AI researcher working across large language model pre-training data, mathematical agents, interactive evaluation, and computational chemistry. Built an end-to-end pre-training data pipeline at Principia Technologies and integrated **AutoMathText-2.5**, to which I contributed, into model-training workflows. Creator and leading developer of **MathCode**, an open-source mathematical coding and formalization agent with 700+ GitHub stars; first author of *Web World Models*; and co-author of work on *Interactive Benchmarks* and *Fast Weight Attention for Continual Learning*. Producing multiple first author peer-reviewed publications in J. Chem. Phys and Phys. Rev. B.

EDUCATION

Aug. 2021–May. 2027 **University of Pennsylvania**, Ph.D. in Chemistry, GPA: 4.0/4.0 — Philadelphia, PA
Advisors: Prof. Joseph E. Subotnik and Prof. Abraham Nitzan

Sept. 2017–Jun. 2021 **University of Science and Technology of China**, B.S. in Physics — Hefei, China

Jul. 2018 **Tsinghua University**, Summer School in Quantum Information — Beijing, China

EXPERIENCES

Researcher | AI for Mathematics / Pre-training Data — Principia Technologies, Inc.
Large-scale pre-training data, pipeline infrastructure, deduplication, tokenization, distributed processing — *Jun. 2026–Aug. 2026*

- Built a complete end-to-end pre-training data pipeline for mathematical and formal-reasoning language models, covering heterogeneous dataset ingestion, normalization, filtering, deduplication, tokenization, and training-ready artifact generation
- Integrated AutoMathText-2.5 and other mathematical and scientific corpora into reproducible pre-training workflows, connecting public dataset development with downstream model training
- Building up evaluation pipeline and benchmark for formalized mathematical proving of models

Researcher | Quantum–Classical Dynamics — Princeton University
Research with Prof. Joseph E. Subotnik; geometric forces, moving-basis effects, chiral molecular systems — *Jul. 2024–present*

- Develop and benchmark classical and mixed quantum–classical dynamics methods, emphasizing electronic-state truncation, geometric forces, moving-basis effects, and conservation laws
- Study electron–nuclear, spin–phonon, and chiral molecular dynamics relevant to chirality-induced spin selectivity and molecular transport

PROJECTS

- **MathCode (`https://github.com/math-ai-org/mathcode`)** — *2026–present*
  - Create and lead the development of an open-source formalization and verified reasoning math-agent; the project has received 700+ GitHub stars
  - Built workflows combining language-model reasoning, code and tool execution, theorem search, Lean 4 verfication and iterative error diagnosis.
  - The project is based on another open-source project of mine which is **AUTOLEAN** (`https://github.com/T3S1AMAX/autolean`), an agentic pipeline for auto formalization and verified math proving
- **Web World Models (`https://princeton-ai2-lab.github.io/Web-World-Models/`)** — *2025*
  - First author of a framework for persistent, controllable, open-ended language-agent environments that combines code-defined world state and dynamics with model-generated context and high-level decisions
- **Interactive Benchmarks (`http://www.interactivebench.org`)** — *2026*
  - Co-developed an evaluation paradigm for active information acquisition and budgeted multi-turn reasoning across logic, mathematics, UI reconstruction, and strategic environments
- **AutoMathText-2.5 (`https://huggingface.co/datasets/math-ai/AutoMathText-2.5`)** — *2026*
  - Contribute to a large-scale mathematical and STEM corpus for language-model pre-training, including data curation, quality filtering, deduplication, and training-data preparation. The dataset monthly receives 12k+ downloads
  - Integrate AutoMathText-2.5 into the complete Principia pre-training pipeline and used the processed data in downstream training workflows
- **Fast Weight Attention for Continual Learning (`https://arxiv.org/abs/2608.27763`)** — *2026*
  - Co-authored research on recurrent fast-weight memories, online continual-learning updates, and efficient sequence modeling
- **An AI-Assisted Meta-Analysis of CISS (`https://jichen-feng.com/posters/ciss-meta-study-poster.pdf`)** — *2025–present*
  - Built an AI-assisted workflow for extracting structured measurements, experimental conditions, and metadata from research literature
  - Applied it to 279 CISS papers and 6,744 measurements; presented preliminary results in the CISS Online Seminars; first-author manuscript in preparation

RESEARCHES

- **Graduate Research: Chirality-Induced Spin Selectivity** — *May. 2023–present*
  Advisors: Prof. Abraham Nitzan, University of Pennsylvania, and Prof. Joseph E. Subotnik, Princeton University
  - Developed chirality descriptors for small-molecule normal modes, including continuous chirality measures and linear/angular-momentum pseudoscalar measures
  - Demonstrated vibrational angular-momentum generation and vibrational energy-transfer rectification in driven chiral molecular systems using classical molecular dynamics
  - Investigated polaron formation and dynamics in chiral molecular systems using mixed quantum–classical dynamics
  - Benchmarked Ehrenfest dynamics and studied Berry-force and moving-basis corrections for momentum and angular-momentum conservation in truncated electronic spaces
  - Conducted an AI-assisted CISS meta-analysis spanning 279 papers and 6,744 datapoints; identified robust within-paper and cross-paper trends
- **Graduate Research: Bulk Photovoltaic Effect** — *May. 2022–May. 2023*
  Advisor: Prof. Andrew M. Rappe, University of Pennsylvania
  - Developed a real-time photocurrent simulation algorithm for solid-state systems
  - Investigated shift-current responses in noncollinear magnetic materials

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PUBLICATIONS

1. **Jichen Feng**, Ethan Abraham, Joseph E. Subotnik, and Abraham Nitzan, "Vibrational angular momentum generation in thermally driven chiral systems," *Physical Review B* **113**, 094301 (2026).
2. **Jichen Feng**, Yifan Zhang, Chenggong Zhang, Yifu Lu, Shilong Liu, and Mengdi Wang, "Web World Models," *arXiv preprint arXiv:2512.23676* (2025).
3. **Jichen Feng**, Ethan Abraham, Joseph E. Subotnik, and Abraham Nitzan, "Rectification of vibrational energy transfer in driven chiral molecules," *The Journal of Chemical Physics* **163**(23) (2025).
4. **Jichen Feng**, Ethan Abraham, Joseph E. Subotnik, and Abraham Nitzan, "Chiral vibrational modes in small molecules," *The Journal of Chemical Physics* **163**(1) (2025).
5. **Jichen Feng**, Joseph E. Subotnik, and Abraham Nitzan, "On the number of states to include in mixed quantum–classical dynamics: Can more states be worse?" manuscript in preparation.
6. **Jichen Feng**, Joseph E. Subotnik, and Abraham Nitzan, "An AI-Assisted Meta-Analysis of Chirality-Induced Spin Selectivity," manuscript in preparation.
7. Aritra Das, **Jichen Feng**, Nazifa Tabassum, Javier Garcia Barriocanal, Abraham Nitzan, Joseph E. Subotnik, Brian Bloom, David H. Waldeck, and Renee Frontiera, "Phonon Angular Momentum Drives Chiral Induced Spin Selectivity," manuscript in preparation.
8. Xuecheng Tao, Clàudia Climent, Ethan Abraham, **Jichen Feng**, and Abraham Nitzan, "Chiral vibrational modes and vibrational circular dichroism," *arXiv preprint arXiv:2607.22257* (2026).
9. Baoqing Yue, Zihan Zhu, Yutong Han, Brian Fan, Qian Sun, **Jichen Feng**, Hufei Yang, Yifan Zhang, and Mengdi Wang, "Interactive Benchmarks," *arXiv preprint arXiv:2603.04737* (2026). Submitted to NeurIPS 2026.
10. Yifan Zhang, Steve Ta, Jasper Zhang, **Jichen Feng**, Shuzhen Li, Yongxin Zhang, Yifeng Liu, Huizhuo Yuan, Mengdi Wang, Quanquan Gu, and Andrew Chi-Chih Yao, "Fast Weight Attention for Continual Learning," *arXiv preprint arXiv:2608.27763* (2026).

TECHNICAL SKILLS

- **LLM Data & Evaluation:** pre-training data pipelines, dataset sourcing and curation, data deduplication, tokenization, agentic evaluation
- **Programming :** Python, Julia, C, MATLAB, Git, Linux, Slurm
- **Scientific Computing:** DFT, electronic-structure calculations, molecular dynamics, mixed quantum–classical dynamics, Quantum ESPRESSO, LAMMPS, GAMESS

SELECTED TALKS & AWARDS

Gordon Research Conference: Molecular Interactions and Dynamics Gordon Research Conference, poster — Jun. 2026
CISS Online Seminars, invited research talk — Jun. 2026
APS Global Physics Summit, research talk — Mar. 2026
Gordon Research Conference: Electron Spin Interactions with Chiral Molecules and Materials, poster — Aug. 2025
Graduate Committee's List, University of Pennsylvania — 2022
Outstanding Student Scholarship, USTC — 2018–2020